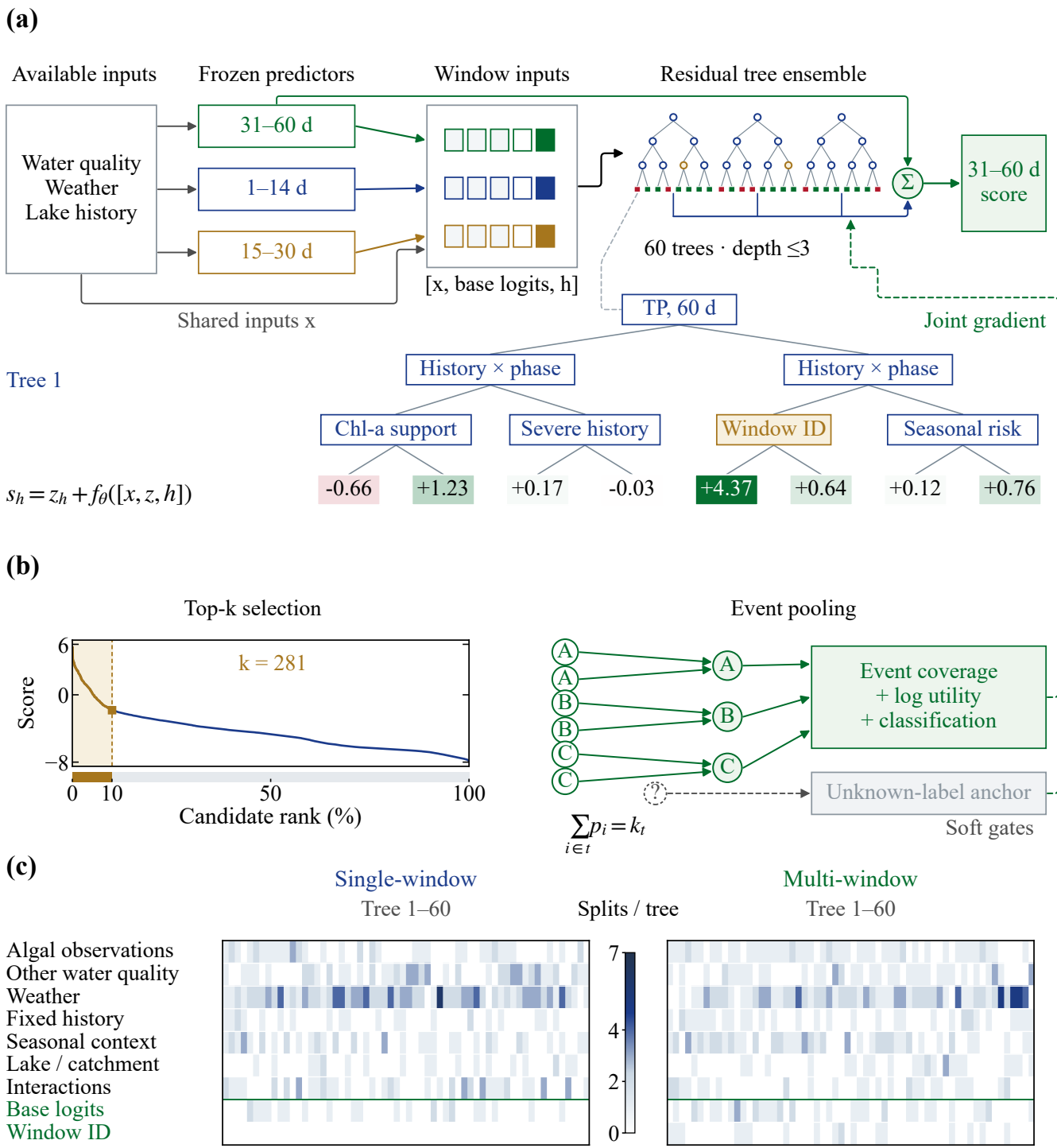

(a)
Available inputs
Frozen predictors
Window inputs
Residual tree ensemble
Water quality
Weather
Lake history
31–60 d
1–14 d
15–30 d
[x, base logits, h]
60 trees · depth ≤3
31–60 d score
Shared inputs x
Joint gradient
Tree 1
TP, 60 d
History × phase
History × phase
Chl-a support
Severe history
Window ID
Seasonal risk
-0.66
+1.23
+0.17
-0.03
+4.37
+0.64
+0.12
+0.76
$s_h = z_h + f_\theta([x, z, h])$
(b)
Top-k selection
Event pooling
k = 281
Score
Candidate rank (%)
Event coverage
+ log utility
+ classification
Unknown-label anchor
Soft gates
$\sum_{i \in t} p_i = k_t$
(c)
Single-window
Tree 1–60
Splits / tree
Multi-window
Tree 1–60
Algal observations
Other water quality
Weather
Fixed history
Seasonal context
Lake / catchment
Interactions
Base logits
Window ID